Augmenting data structures exercises

14-E01 Select and size

Difficulty: Basic

x has left size5, itself1, and right size7. To find the eighth smallest in its subtree, choose a direction and the new k.

14-E02 Updating size on rotation

Difficulty: Advanced

Left-rotate x so right child y rises and its left subtree beta moves to x. Give the recomputation order for x.size and y.size.

14-E03 Interval overlap search

Difficulty: Advanced

An interval-tree root stores[20,30], with left.max25 and right.max34. How do max fields guide queries[24,26] and[31,33]?

14-E04 Select and rank in the same tree

Difficulty: Advanced

Use the lecture tree20,10,40,5,15,30,50,25,35,27. Trace select7 and rank(node27). Delete20 by transplanting25 to root; give size30,size40,and new root25.size.

14-E05 Local formulas versus global-rank maintenance

Difficulty: Research

Compare the numbers of affected nodes when inserting a new minimum with explicitly stored global ranks versus subtree sizes. Does a child-computable summary always need only two updates per rotation?

14-E06 A feasible maximum does not guarantee overlap

Difficulty: Research

A closed-interval BST has root[20,21],left[10,40],right[30,35]. Query[5,6]. Does left.max≥q.low hold? Is there overlap? Use the left-descent proof to explain why searching right afterward is unnecessary.

> Complete your derivations before opening the separate solutions PDF.

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Augmenting data structures exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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